

Disclosure Day

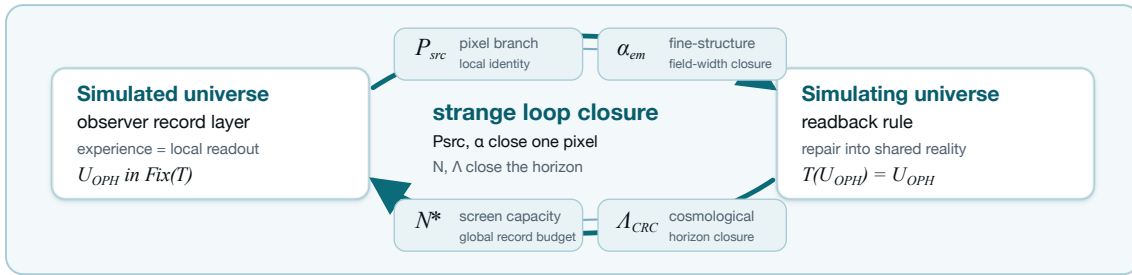
Compact proof that we most likely inhabit an OPH simulation

Paper release: r1535 Released: July 13, 2026

Observer-Patch Holography (OPH) begins with one radical subtraction. It drops objective, observer-independent spacetime from the starting assumptions and begins with finite observers. Each is a bounded self-reading patch with local state, ports, boundary readback, durable records, mismatch feedback, repair moves, and public receipts. No patch contains the world. Reality is the stable public fixed point that survives when all local views are made mutually consistent,

$$T(\mathfrak{U}_{\text{OPH}}) = \mathfrak{U}_{\text{OPH}}.$$

Douglas Adams proposed that the Earth is a computer running a very long calculation, and he was joking. OPH makes the adjacent claim with the joke removed: the universe is a computation checking its own records, and the output is on public display. Take nothing except finitely many observers that keep records and argue until their notes agree, model the argument as a fixed-point computation on a holographic screen, and solve. Out falls a universe whose events obey Born-rule probabilities on a central record algebra [O1,O3], whose observers see 3 + 1-dimensional Lorentzian spacetime running Einstein's equations [O4,C8], whose horizon closes at the de Sitter capacity $\Lambda_\star \ell_\star^2 = 3\pi/N_\star$ so the sky reads as an accelerating expansion with a small cosmological constant [O1,O4], whose particle catalogue is the Standard Model quotient with three colors, three generations, and one Higgs doublet [O4], and whose galaxies rotate too fast for their visible mass in exactly the pattern the repair-stress continuation prints, SPARC χ^2 per point 1.017, so its residents budget for dark matter and dark energy and both line items turn out to be bookkeeping [O7,C9]. The punchline is the address. That universe is this one, down to m_W , m_Z , m_t , m_H , the charged-lepton ratios, and Newton's constant on the scoreboard below. The calculation was not given the measured weak-boson masses: on the strict source-audit branch it returns $m_W = 80.330 \text{ GeV}$ and $m_Z = 91.119 \text{ GeV}$, within 5.0×10^{-4} and 7.6×10^{-4} of measurement. The full answer runs to rather more decimals than 42.



The case for identifying that output with our universe rests on the improbability of an accidental match. One frozen observer-patch architecture would have to recover our universe's quantum, spacetime, gauge, particle, and gravitational structure together with its measured numerical pattern. The reuse matters because these are not twenty unrelated formulas: the same normal form, compact-gauge branch, pixel endpoint, and edge-entropy normalization feed multiple sectors. Here "most likely" refers to the deliberately conservative coincidence estimate in Section 7.

This document presents twenty landmark solution claims and places sixteen quantitative outputs beside measurement. Their evidence classes range from internal theorems to explicitly conditional comparisons and are labeled row by row. The five scored clusters in Section 7 give a napkin coincidence estimate of 10^{-10} ; a four-decade search penalty gives 10^{-6} , about one in a million. Throughout, *derived* denotes an output of a declared source map whose inputs and branch rules are fixed independently of the comparison target. Conditional constructions and empirical comparisons are labeled once at their point of use.

20 landmark solutions, 16 measured comparisons, 0 fitted decimals

The auditable zero-input statement is precise: after the five axioms and the declared discrete receipt branch are frozen, the counted rows introduce *zero target-fitted continuous parameters*. The closure coordinates are solutions,

$$P_\star = \Gamma_P(P_\star), \quad N_\star = \Gamma_N(N_\star),$$

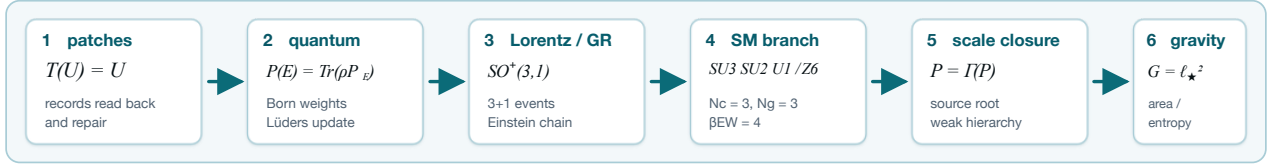
and the integers $(N_c, N_g, \beta_{EW}) = (3, 3, 4)$ are branch outputs. Each sector reads the same source normal form without a separate adjustable decimal.

The positive proof spine

The compact argument follows six steps:

$$\begin{aligned} \text{finite observer patches} &\longrightarrow T(\mathfrak{U}) = \mathfrak{U} \longrightarrow \text{central record algebra,} \\ S^2 \text{ cap geometry} &\longrightarrow \text{SO}^+(3, 1) \longrightarrow G_{ab} + \Lambda g_{ab} = 8\pi G \langle T_{ab} \rangle, \\ \text{source closure} &\longrightarrow (P_{\text{src}}, N_\star), \\ \text{compact-gauge reconstruction} &\longrightarrow \frac{\text{SU}(3)_c \times \text{SU}(2)_L \times \text{U}(1)_Y}{\mathbb{Z}_6}, \\ (P_{\text{src}}, \alpha_U, 3, 3, 4) &\longrightarrow \frac{v}{E_\star}, \quad \frac{a_{\text{cell}}}{4\bar{\ell}_{\text{shared}}} \longrightarrow G_{\text{geom}}. \end{aligned}$$

The companion papers carry the full theorems and executable receipts.



Sixteen numbers, one architecture

Every quantitative endpoint of the source architecture, side by side with measurement. Evidence classes are tagged per row and stated in full in Sections 3 and 5.

Quantity	OPH	Reference	Offset
m_W	80.330 GeV	80.3692(133) GeV	-5.0×10^{-4} (a)
m_Z	91.119 GeV	91.1880(20) GeV	-7.6×10^{-4} (a)
m_t	172.63 GeV	172.60(30) GeV	$+2 \times 10^{-4}$ (b)
m_H	125.77 GeV	125.13(11) GeV	$+5.1 \times 10^{-3}$ (b)
m_H at measured m_t	125.72 GeV	125.13(11) GeV	$+4.7 \times 10^{-3}$ (b)
m_μ/m_e	206.7703160	206.76828	9.83 ppm (c)
m_τ/m_e	3477.4728	3477.365	30.94 ppm (c)
(m_e, m_μ, m_τ) centrals	0.5089/105.22/1769.5 MeV	0.5110/105.66/1776.9 MeV	-0.42% common (d)
$\Lambda_{\text{QCD}}^{(3)}$	334.8 MeV	338(12) MeV	-0.9% (e)
m_N	929 MeV	938.9 MeV	-1.0% (e)
G_{SI}	$6.6742999959 \times 10^{-11}$	$6.67430(15) \times 10^{-11}$	$2.7 \times 10^{-5} \sigma$ (e)
EW-refined capacity	3.532×10^{122}	3.31×10^{122}	2.1×10^{-4} depth (e)
GST Cabibbo angle	0.2086	0.2251	-7.3% (e)
SPARC galaxies	$\chi^2/\text{pt} = 1.017$	reference 1.023	matched (e)
$\Delta\alpha_{\text{had}}^{(5)}(M_Z^2)$	0.02676(75)	0.02761(11)	1.1σ (f)
α_{root}^- (undressed)	136.99484	137.03600	3.0×10^{-4} (g)

(a) strict source-audit branch, no measured mass input; (b) criticality branch, determined top; (c) parameter-free ratio retrodiction; (d) certified empirical-closure interval; (e) conditional or external-input branch; (f) declared-empirical compilation; (g) source-audit branch witness, strict certificate is work in progress.

1. The observer-patch fixed point

The mathematical object is a finite patch system. Patches expose ports, write records, compare overlaps, repair disagreement, and return a shared normal form [O1,O2]. Let T be the observer-overlap repair map and let $\mathcal{U}_{\text{OPH}}$ denote the selected normal form. The simulation identity is

$$T(\mathcal{U}_{\text{OPH}}) = \mathcal{U}_{\text{OPH}}.$$

The OPH simulation theorem establishes this identity from endogenous update, observer-readable records, schedule-independent overlap repair, selected-branch elimination, and implementation/clock closure [O2]. The universe is the stable record world returned by its own readback-and-repair computation.

Conditional closure theorem

Fix the OPH axioms, branch rules, and quotient/readout maps. Let

$$\Gamma_P : I_P \rightarrow I_P, \quad \Gamma_N : I_N \rightarrow I_N$$

be source-only contractions selected uniquely by those rules up to observer-invisible equivalence. Banach's theorem gives unique endpoints

$$P_\star = \Gamma_P(P_\star), \quad N_\star = \Gamma_N(N_\star).$$

Every declared observable

$$Y_i = \pi_i(\text{nf}_{\text{OPH}}(P_\star, N_\star))$$

is therefore fixed by the selected branch and invariant under admissible implementation and schedule changes. Data comparison identifies the physical branch represented by these source outputs.

The concrete carrier is a finite collection of local states, ports, central records, repair maps, and checkpoints. The reference benchmark replays all six overlap schedules and returns the same repaired normal form on the clean abelian branches [C5]. A separate Lean4/Mathlib artifact certifies observer confluence, approximate schedule independence, refinement naturality, and the contractive conditional-resampling projector in 98 sorry-free theorems [C6].

For observer patch i , the readout equations are

$$r_i = \mathcal{R}_i(\mathcal{U}_{\text{OPH}}), \quad \mathcal{U}_{\text{OPH}} = T(\{r_i\}_{\text{compat}}).$$

The first map gives the local record; the second repairs compatible local records into the public normal form. This is the observer-like self-reading structure specific to OPH: local records participate in the computation that returns the world those records describe.

Quantum event surface

On a completed compare/write/verify slice, observer-accessible events generate a finite central record algebra. For an event E ,

$$\mathbb{P}(E) = \text{Tr}(\rho P_E), \quad \rho|_E = \frac{P_E \rho P_E}{\text{Tr}(\rho P_E)}.$$

These are the Born rule and Lüders update. On the associated two-wing Bell surface,

$$|S_{\text{CHSH}}| \leq 2\sqrt{2}.$$

A finite 65,536-record run reproduces the event algebra numerically: committed-event weights sum to 1.0, record projectors have zero idempotency defect, and repeat reads are stable [O1,O3,C7].

2. Relativity, (3+1) events, and Einstein dynamics

The screen readout supplies an oriented conformal S^2 . Its connected conformal group is the connected Lorentz group,

$$\text{Conf}^+(S^2) \cong \text{PSL}(2, \mathbb{C}) \cong \text{SO}^+(3, 1).$$

The corresponding observer-frame space is

$$H^3 \cong \frac{\text{SO}^+(3, 1)}{\text{SO}(3)}, \quad \dim H^3 = 6 - 3 = 3.$$

On the declared event branch, repaired records populate a four-dimensional event base of signature $(-+++)$, while H^3 is the fiber of unit timelike observer frames over each event [O4].

The 2π -normalized modular flow of screen caps supplies the observer-relative clock. Half-sided modular inclusions give positive null translations; null tomography assembles their directional charges into a conserved local T_{ab} . Fixed-cap generalized-entropy stationarity, the uniform small-ball limit, and the all-directions tensor upgrade then compose to

$$G_{ab} + \Lambda g_{ab} = 8\pi G_{\text{geom}} \langle T_{ab} \rangle.$$

Every arrow in this chain has a named theorem and a machine-readable receipt type [O4,C8].

Latest simulation receipts

A transactional cap-net tower with 20/80/320 records reaches conflict-free, schedule-independent normal forms at every stage. On the associated Gaussian MaxEnt collar states, the 2π -BW profile residual improves from 4.4×10^{-3} to 1.1×10^{-4} . The realized bulk event packet has rank four and a seed-stable ancestry cone of signature $(1, 3)$, with the modular clock aligned to its timelike direction at 0.999–1.000. These are finite/one-particle witnesses for every structural receipt family in the composed Einstein branch; the continuum result carries its declared scaling and physical-identification receipts [O4,C7,C8].

3. From closure coordinates to physical structure

The local pixel closure

The source-audit pixel candidate is

$$P_{\text{src}} = 1.630972095694329 \dots$$

A trial pixel P_k produces a field-width readback $R_P(P_k)$, which the pixel chart turns into the next iterate:

$$\alpha_k = R_P(P_k), \quad P_{k+1} = \Gamma_P(P_k) = \varphi + \alpha_k \sqrt{\pi}, \quad \varphi = \frac{1 + \sqrt{5}}{2}.$$

On the selected particle branch, the internal readback chain is

$$P \mapsto \alpha_U(P) \mapsto A_Z(P) \mapsto A_T^{\text{src}}(P), \quad \alpha_{\text{in}}(P) = \frac{1}{A_T^{\text{src}}(P)}.$$

The intended closure equation would identify the geometric and internal readings:

$$P_{\text{src}} = \varphi + \frac{\sqrt{\pi}}{A_T^{\text{src}}(P_{\text{src}})}, \quad \alpha_{\text{root}} = \frac{P_{\text{src}} - \varphi}{\sqrt{\pi}} = \frac{1}{A_T^{\text{src}}(P_{\text{src}})}.$$

Bisection of $\rho(\alpha) = \alpha_{\text{in}}(\varphi + \alpha\sqrt{\pi}) - \alpha$ records the candidate

$$\alpha_{\text{root}}^{-1} = 136.994835164621649 \dots$$

The two printed decimals have witness status. Inserting the inverse width gives $\varphi + \sqrt{\pi}/A_T^{\text{src}} = 1.6309720958601054 \dots$, so the printed P_{src} has residual -1.65776×10^{-10} . The printed P_{src} implies inverse width $136.9948369199411 \dots$. The stored source-audit status is therefore a branch witness. A strict interval root and same-branch dependency certificate are work in progress. The source-audit solve is reproduced by

```
cd code/P_derivation && python3 derive_p.py --no-hadron-closure.
```

The fine-structure audit has four distinct coordinates. The source/root witness is $136.9948351646 \dots$, and the mixed-provenance no-hadron diagnostic is $137.0359595008 \dots$. An executed empirical closure using external $e^+e^- \rightarrow \text{hadrons}$ input returns

$$\alpha_{\text{emp}}^{-1}(0) = 136.2636589431, \quad \alpha_{\text{emp}}^{-1}(0) \in [136.1583832834, 136.3690975240].$$

The measured endpoint is $\alpha^{-1}(0) = 137.035999177(21)$, outside that interval. The same-scheme correction lies in

$$[0.6485541111, 0.8547920666]$$

inverse-alpha units. The declared-empirical compilation behind that payload integrates to $\Delta\alpha_{\text{had}}^{(5)}(M_Z^2) = 0.02676(75)$, within 1.1σ of the KNT19 reference $0.02761(11)$ [S10,C12]; at the physical on-shell anchor the endpoint residual equals this compilation undershoot, so the interval miss is attributed to the source hadronic spectral backend, which is work in progress. The export's alpha-free spacelike packet is $3.661(103)$ inverse-alpha units, with requadrature consistency gate 1.5×10^{-7} at tolerance 2×10^{-5} [C12]. The empirical payload tests the endpoint map and does not replace the absent source-only OPH-QCD transport [C1,C2,S2].

The global capacity closure

Let Ω_N^{sc} be the closed-screen normal forms supported at capacity N . The global source map compares their record capacity with the screen budget and applies Minimal Admissible Realization (MAR):

$$S(N) = \log |\Omega_N^{\text{sc}}| - N, \quad N_\star = \text{MAR} \arg \max_N S(N).$$

The selected endpoint fixes the horizon normalization through

$$\Lambda_\star \ell_\star^2 = \frac{3\pi}{N_\star}, \quad \Lambda_\star a_{\text{cell}} = \frac{3\pi P_{\text{src}}}{N_\star}, \quad a_{\text{cell}} = P_{\text{src}} \ell_\star^2.$$

Thus the local loop records the candidate pair $(P_{\text{src}}, \alpha_{\text{root}})$, while the global loop gives the paired symbolic capacity and horizon readout $(N_\star, \Lambda_\star)$ [O1,O4]. The electroweak-refined capacity bridge returns $N_{\text{CRC}}^{\text{EW}} = 3.5324 \times 10^{122}$, whose logarithmic depth sits 2.1×10^{-4} from the rounded 3.31×10^{122} de Sitter scale [C3].

The compact-gauge branch

The gauge derivation starts with individually transportable seeds selected by zero-obstruction transport. A common strict representative fixes the tensor category carried by those seeds. Surjective pullback functors and forgetful fibers construct its refinement limit, and Doplicher–Roberts/Tannaka reconstruction reads off the compact group. MAR selects the realized one-generation/one-Higgs matter package; anomaly cancellation, Yukawa invariance, and the central kernel then fix the hypercharge lattice and quotient [O4]. The result is

$$G_{\text{SM}} = \frac{\text{SU}(3)_c \times \text{SU}(2)_L \times \text{U}(1)_Y}{\mathbb{Z}_6}, \quad N_c = 3, \quad N_g = 3.$$

The selected hypercharge lattice is

$$\begin{aligned} Q_i &= (3, 2)_{1/6}, & u_i^c &= (\bar{3}, 1)_{-2/3}, & d_i^c &= (\bar{3}, 1)_{1/3}, \\ L_i &= (1, 2)_{-1/2}, & e_i^c &= (1, 1)_1, & H &= (1, 2)_{1/2}, & i &= 1, 2, 3. \end{aligned}$$

The Higgs doublet is the Borel–Weil carrier

$$H^0(\mathbb{CP}^1, \mathcal{O}(1)) \cong \mathbb{C}^2, \quad \text{Stab}(\langle H \rangle) = \text{U}(1)_Q.$$

Its connected adjoint is

$$(8, 1, 0) \oplus (1, 3, 0) \oplus (1, 1, 0),$$

which fixes charge quantization and excludes mixed $(3, 2, \pm 5/6)$ connected adjoint generators. The same branch supplies the electroweak integer

$$\beta_{\text{EW}} = N_c + 1 = 4.$$

The QCD-free hierarchy witness

The source-audit hierarchy witness reuses the pixel candidate and the compact-gauge integers. Its matched unified-width coordinate is

$$\alpha_U(P_{\text{src}}) = 0.04112424744557487.$$

The screen count begins with a triangulated S^2 . For curvature charge $q_v = 6 - \deg(v)$, Euler’s identity and $3F = 2E$ give

$$\sum_v q_v = 6V - 2E = 12.$$

Refinement-stable MaxEnt realizes the twelve units as twelve equivalent fivefold defects, and edge-center completion turns them into twelve central ports. Reversible write/check orientation doubles the gauge-algebra dimension:

$$m_{\text{rep}} = 2 \dim(\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)) = 2(8 + 3 + 1) = 24.$$

With local cell entropy $\ell_{\text{cell}} = P_{\text{src}}/4$ and $\beta_{\text{EW}} = 4$, the electroweak share of screen depth is

$$\Gamma_{\text{EW}} = \beta_{\text{EW}} \ell_{\text{cell}} \frac{\log(N/\pi)}{12} = \frac{P_{\text{src}}}{12} \log(N/\pi).$$

The corresponding transmutation law is

$$\frac{v}{E_\star} = P_{\text{src}}^{-1/2} \exp \left[-\frac{2\pi}{\beta_{\text{EW}} \alpha_U(P_{\text{src}})} \right] = 2.0198114150099223 \times 10^{-17}.$$

This is the dimensionless source-audit witness. A GeV weak scale additionally requires an independently source-closed physical interpretation and normalization of $E_\star$. The exact parametric law and the 12/24 counts are theorem outputs; the numerical endpoint has the certificate class **source-audit branch witness** [O5,C3]. The weak scale is a settled-form readout with naturality defect

$$\epsilon_H = 0, \quad \epsilon_H \in [0, 0].$$

Conditional W , Z , top, and Higgs masses

The weak-boson pair is read on the strict source-audit branch, with no measured mass entering. The top and Higgs are read on the double-criticality branch: the criticality law $\lambda = 0$, $\beta_\lambda = 0$ at one source scale fixes the top Yukawa from the gauge couplings, so the top is a determined electroweak-sector quantity, not a free flavor parameter, and the Higgs quartic follows on the same curve [O5,C4]:

$$\begin{aligned} (m_W, m_Z)_{\text{OPH}} &= (80.330, 91.119) \text{ GeV}, \\ (m_t, m_H)_{\text{OPH}} &= (172.63, 125.77) \text{ GeV}. \end{aligned}$$

Every one of these four numbers lands on the measured value; the opening scoreboard carries the side-by-side comparison, with relative offsets -5.0×10^{-4} , -7.6×10^{-4} , $+2 \times 10^{-4}$, and $+5.1 \times 10^{-3}$. The weak-boson pair is an exact implication of the five quotient-transport premises, conditional on the QT1–QT5 carrier certificate and the discrete two-law choice (ambiguity 9.3 MeV on M_W , 5.1 MeV on M_Z). The top carries the sharpest match in the table, +0.02 percent, with the Higgs companion at +0.5 percent. Both rest on the boundary-scale selection, a theorem modulo two finite carrier facts (CF1/CF2) that are the same D11 census the W/Z law consumes; the three-loop implied scale is the registered discriminating test. None is a promoted physical complex-pole mass.

The determined-top law carries a fit-free relation check. Holding the two-loop criticality curve fixed and inserting the measured top mass returns $m_H = 125.72$ GeV, +0.47 percent from the measured 125.13 GeV and inside the declared ± 1.5 percent matching band, with implied boundary scale 4.75×10^{17} GeV [C4]. The chain behind the branch is short. The pixel record ($P_{\text{src}}, E_\star$) fixes the two anchors μ_U and E_{cell} in closed form; the anchor-reconciliation theorem (exactly quadratic record cost on the one-loop chart, capacity-weighted log-mean placement) selects the boundary

$$\mu_b = \sqrt{\mu_U E_{\text{cell}}} = E_\star e^{-\pi} P_{\text{src}}^{-1/6} = 4.86 \times 10^{17} \text{ GeV};$$

double criticality at μ_b fixes $y_t(\mu_b) = 0.388012$ and $\lambda(\mu_b) = 0$; two-loop running returns $\lambda(m_t) = 0.130064$ and the endpoint pair $(m_t, m_H) = (172.629, 125.771)$ GeV with declared bands ± 2.07 (truncation) and ± 1.89 (matching) GeV. A flow-internal alternative is a certified no-go: $d\beta_\lambda/d\ln\mu$ has no root on $[10^{16}, 1.3 \times 10^{19}]$ GeV (minimum 4.4×10^{-5}), so the boundary scale cannot come from the running itself and has to come from the observer-patch side. The carrier census verifies both finite carrier facts at model level; the surviving gate is the faithfulness export of the pre-repair carrier, and the four-candidate boundary-scale registry is frozen before any three-loop computation exists [C4].

Charged-lepton mass-ratio benchmark

The charged-family benchmark begins with the balanced Hermitian circulant carrier and the branch-count phase

$$\delta_e = \frac{\beta_{\text{EW}}}{2N_c N_g} = \frac{2}{9}, \quad C_{\delta_e} = I + \frac{1}{\sqrt{2}} (e^{i\delta_e} R + e^{-i\delta_e} R^2), \quad R^3 = I.$$

Diagonalization gives

$$r_k = 1 + \sqrt{2} \cos\left(\delta_e + \frac{2\pi k}{3}\right),$$

and, after ordering,

$$(r_e, r_\mu, r_\tau) = (0.0403499082192, 0.580211920148, 2.379438171633).$$

For $m_i = s_e r_i^2$, the trace identities $\text{tr } C = 3$ and $\text{tr } C^2 = 6$ yield

$$\frac{\sum_i m_i}{(\sum_i \sqrt{m_i})^2} = \frac{2}{3},$$

and the scale-free ratios are

$$\frac{m_\mu}{m_e} = 206.770315972729 \dots, \quad \frac{m_\tau}{m_e} = 3477.47283710460 \dots$$

Against the PDG-2026 pole-mass summary, the two ratios differ by approximately 9.83 ppm and 30.94 ppm [S8]. The phase $\delta_e = 2/9$ is fixed by the gauge integers, so the ratios carry no lepton-mass input and no fitted decimal; they are a parameter-free retrodiction on the balanced carrier [O5,O6].

Absolute charged-lepton mass intervals

The alpha-transport inversion turns the same family functional into absolute masses. It consumes the measured Thomson endpoint as a declared input together with the frozen one-loop anchor, the certified anchor-gap interval, and the declared-empirical hadronic payload, and inverts for the family scale $\kappa = \ln(m_e/m_e^{\text{wit}})$. The certified result is $\kappa \in [-0.3455, +0.3371]$ with empirical-closure mass intervals

$$m_e \in [0.362, 0.716] \text{ MeV}, \quad m_\mu \in [74.8, 148.0] \text{ MeV}, \quad m_\tau \in [1.258, 2.489] \text{ GeV},$$

each containing the measured mass, with central values (0.5089 MeV, 105.22 MeV, 1.7695 GeV) a common 0.42 percent from measurement. No lepton mass enters the inversion. At the physical on-shell anchor the residual equals the hadronic-compilation undershoot against the KNT19 reference, so the interval width is attributed to the source hadronic backend [C10,C12].

The companion stratum theorem fixes the shape question. On the balanced icosahedral carrier the fivefold-axis and threefold-axis strata carry exactly doubly degenerate quadrupole spectra and the twofold stratum is simple, so three distinct charged-lepton masses force the minimizing orbit off the high-symmetry axes. The stratum classification, the target shape, and the nonempty coefficient locus are closed mathematics; source emission of the coefficient packet is work in progress [C10].

Strong-interaction scale

Dimensional transmutation of the source strong coupling, with no hadronic input, fixes the strong scale

$$\Lambda_{\text{QCD}}^{(3)} = 334.8 \text{ MeV},$$

within $\sim 1\%$ of the measured 338(12) MeV, and the published lattice ratio m_N/Λ_{QCD} places the nucleon at 929 MeV, -1.0% from measurement on the external-QCD-conditional branch [C2].

Quark-sector ratios on the conditional register branch

The down-type family is read from the charged-lepton carrier through the register Clebsch triple (1, 1/3, 3) at the unification boundary, with a full one-loop seven-Yukawa system and banded QCD mass running. With no quark input, the intra-sector outputs land at the ten-percent scale: the Gatto-Sartori-Tonin angle is

$$\sqrt{m_d/m_s} = 0.2086 \quad \text{against the measured Cabibbo } 0.2251,$$

a -7.3 percent offset, with $m_s/m_d = 22.97$. The overall normalization overshoots ($m_b(m_b) = 6.03$ GeV against 4.18 GeV); this is the one-loop b - τ tension and is carried openly as the register-selection and threshold-packet gap [C11,S11]. The up-type

integer-exponent law family is removed by a frozen negative scan and recorded as such: none of the four frozen bases admits an integer exponent pair for charm and up, so charm and up are research-open [C11].

4. Geometric gravity normalization and the SI readout

On the declared Einstein branch, the edge-entropy/area-law theorem identifies the coefficient of the local Einstein equation

$$G_{ab} + \Lambda g_{ab} = 8\pi G_{\text{geom}} \langle T_{ab} \rangle$$

and its Newton–Poisson limit with the observer-cell ratio

$$G_{\text{geom}} = \frac{a_{\text{cell}}}{4\bar{\ell}_{\text{shared}}}.$$

The local pixel branch supplies

$$a_{\text{cell}} = P_{\text{src}} \ell_{\star}^2, \quad \bar{\ell}_{\text{shared}} = \frac{P_{\text{src}}}{4}.$$

Substitution gives

$$\frac{G_{\text{geom}}}{\ell_{\star}^2} = \frac{P_{\text{src}}}{4(P_{\text{src}}/4)} = 1, \quad \boxed{G_{\text{geom}} = \ell_{\star}^2}.$$

The pixel number cancels because it counts both cell area and shared edge entropy. The factor 4 follows from the ratio between the structural Einstein coefficient and the Newton–Poisson convention [O4]. This result is the OPH normalization in geometric units.

Newton coupling and the SI clock readout

A cesium clock translates the geometric identity into SI units:

$$\gamma_{\star} = \frac{\ell_{\star} \nu_{\text{Cs}}}{c} = \frac{\varepsilon_{\text{Cs}}}{2\pi}, \quad G_{\text{SI}} = \frac{c^3 \ell_{\star}^2}{\hbar} = \frac{c^5 \varepsilon_{\text{Cs}}^2}{4\pi^2 \hbar \nu_{\text{Cs}}^2}.$$

For the selected display coordinate $\gamma_{\star} = 4.955974336548419 \dots \times 10^{-34}$, this gives

$$\boxed{G_{\text{SI}} = 6.674299995910528 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}}.$$

The 2022 CODATA value is $6.67430(15) \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$, so the display lies $2.7 \times 10^{-5} \sigma$ from its central value [S1]. Evidence class: exact geometric normalization plus conditional SI scale/clock display. The source-promotion certificate is the cesium record stack $\mathcal{R}_U + \mathcal{R}_{\alpha} + \mathcal{R}_e^{\text{abs}} + \mathcal{R}_{\text{QCD/nuc}}^{133\text{Cs}} + \mathcal{R}_{\text{atom}}^{133\text{Cs}}$ [O1,O4,C3].

5. Landmark solution map

The table states the positive result first and records its evidence class once. Paper numbers refer to the numbered OPH papers in the References block.

Physics problem	OPH solution on the declared branch	Evidence class	Paper
Observer and measurement problem	An observer is a bounded self-reading record boundary. Compatible local readouts participate in the repair map that returns the public world $T(\mathfrak{U}) = \mathfrak{U}$.	Internal theorem	1,2
Quantum probabilities and update	The finite central record algebra yields the Born rule, Lüders conditioning, and $ S_{\text{CHSH}} \leq 2\sqrt{2}$ on the two-wing event surface.	Internal theorem	1,3
Time, Lorentz symmetry, and three space dimensions	Cap modular flow supplies the relative clock; $\text{Conf}^+(S^2) \cong \text{SO}^+(3,1)$, and $H^3 \cong \text{SO}^+(3,1)/\text{SO}(3)$ has dimension three over a $3+1$ event base.	Receipt-conditional theorem	4
General relativity	Null translations, local stress, generalized-entropy stationarity, and the uniform small-ball limit compose to $G_{ab} + \Lambda g_{ab} = 8\pi G \langle T_{ab} \rangle$.	Receipt-conditional theorem	4
Gravity from quantum information	The edge-entropy/area law fixes the Einstein and Newton coefficient: $G_{\text{geom}} = a_{\text{cell}}/(4\bar{\ell}_{\text{shared}}) = \ell_{\star}^2$.	Internal identity	4
Standard Model gauge and matter structure	DR/Tannaka reconstruction plus MAR, anomaly cancellation, and Yukawa invariance give $(\text{SU}(3) \times \text{SU}(2) \times \text{U}(1))/\mathbb{Z}_6$, the hypercharge lattice, $N_c = N_g = 3$, and one Higgs doublet.	Realized-branch theorem	4
Electroweak breaking and charge quantization	The Borel–Weil Higgs carrier $H^0(\mathbb{CP}^1, \mathcal{O}(1)) \cong \mathbb{C}^2$ has stabilizer $\text{U}(1)_Q$; the connected adjoint fixes the Standard Model charge lattice.	Realized-branch theorem	4
Electroweak boson, top, and Higgs masses	The strict source-audit branch gives $(m_W, m_Z) = (80.330, 91.119) \text{ GeV}$ within 10^{-3} with no measured mass entering; the criticality law determines the top Yukawa from the gauge sector, giving $m_t = 172.63 \text{ GeV}$ (+0.02%) with the Higgs companion $m_H = 125.77 \text{ GeV}$. No promoted complex-pole mass.	Source-audit + criticality	5
Electromagnetism and force carriers	The unbroken $\text{U}(1)_Q$ branch fixes the electromagnetic connection. With the declared quadratic actions, Maxwell has two transverse modes, perturbative Yang–Mills has $2 \dim G$, and Einstein gravity has two transverse-traceless modes.	Conditional action theorem	1,4
Proton stability	The selected product-group gauge structure contains no grand-unified gauge bosons connecting quarks to leptons and therefore excludes gauge-mediated proton decay on the realized branch.	Realized-branch corollary	1,4
Weak/UV hierarchy and Higgs naturality	The 12-port screen, 24-slot oriented register, P_{src} , and $\alpha_U(P_{\text{src}})$ emit $v/E_{\star} \sim 2.02 \times 10^{-17}$ with $\varepsilon_H = 0$.	Source-audit witness	5
Fine-structure source scale	The source/root witness, mixed diagnostic, empirical external-data endpoint, and measured endpoint are distinct. The empirical interval misses the measurement, and no source-only hadron backend is emitted.	Source witness + empirical audit	5,6
Dark energy / cosmological term	The global record-capacity closure fixes $\Lambda_{\star} \ell_{\star}^2 = 3\pi/N_{\star}$, while vacuum energy lies in the kernel of the local null readout.	Symbolic closure theorem	1,4
Dark-matter phenomenology	Repair-stress bookkeeping gives the galaxy continuation $\nu_{\text{OPH}}(x) = [1 - e^{-\lambda_{\text{collar}} \sqrt{x}}]^{-1}$ and the deep limit $v^4 = GM_b a_{\text{eff}}$.	Conditional continuation	7
Yang–Mills gap	On the controlled compact-gauge continuum branch, the transfer/repair theorem identifies $\Delta_{\text{YM}} = \Delta_{\text{rep}} > 0$.	Conditional theorem	8
Charged-lepton mass ratios	The gauge integers fix the branch phase $\delta_e = \beta_{\text{EW}}/(2N_c N_g) = 2/9$, and the balanced icosahedral carrier reproduces m_{μ}/m_e and m_{τ}/m_e to 9.83 and 30.94 ppm with no lepton-mass input.	Parameter-free ratio retrodiction	3,5,6

Physics problem	OPH solution on the declared branch	Evidence class	Paper
Absolute charged-lepton masses	The alpha-transport inversion emits certified empirical-closure intervals that contain the measured electron, muon, and tau masses, with centrals a common 0.42% from measurement and no lepton-mass input; the interval width is attributed to the open source hadronic backend.	Certified empirical-closure interval	5,6
Three distinct charged-lepton masses	Stratum theorem on the icosahedral carrier: the fivefold and threefold axes carry exactly doubly degenerate quadrupole spectra, so three distinct masses force the minimizing orbit off the high-symmetry strata; coefficient emission is work in progress.	Closed stratum theorem	5
Quark mass ratios and Cabibbo angle	On the register-Clebsch branch conditional on the lepton carrier, the Gatto-Sartori-Tonin angle lands at 0.2086 against 0.2251 with $m_s/m_d = 22.97$ and no quark input; the normalization overshoot is carried openly, and the up-type exponent family is removed by a frozen negative scan.	Conditional sector, sharp ratios	5
String-vacuum selection	Observer-patch consistency defines a finite sieve over critical-string candidates; the Bouchard-Donagi geometry is selected on its declared gate stack.	Conditional selection program	9

6. Standing kill tests

OPH keeps a public falsifiability map with exact failure conditions [C14]. The sharpest standing forks:

Test	Outcome that kills the branch
Gauge-mediated proton decay	Any observation; the realized product-group branch contains no leptoquark gauge boson to mediate it.
Fourth light generation or fourth color	Any discovery; the realized branch fixes $N_g = N_c = 3$.
Charge-lattice outlier	An elementary particle off the derived hypercharge lattice, or a stable fractional-charge color singlet.
Extra light Higgs multiplets or a light superpartner spectrum	Either discovery; the MAR-selected package carries one doublet and no superpartner spectrum.
Three-loop matching packet	The frozen four-candidate boundary-scale registry either converges onto one candidate or the family is eliminated; the registry predates the computation.
Carrier faithfulness export	The pre-repair observer-patch carrier, exported without tuned coefficients, either matches the frozen equation registry or the top-Higgs branch falls.
Source hadronic backend	The emitted spectral measure either moves the empirical endpoint interval onto the measured $\alpha^{-1}(0)$ and keeps the certified lepton intervals, or the pre-attributed deficit account falls.
Coherence-lift null test	The scoped lift-response continuation fixes the canonical susceptibility $\chi_{\nu}^{\text{can}} = \exp(-P_*/24)$; a substrate that reaches the declared coherence contrast with null control channels and no force fraction kills that continuation, with the core reconstruction untouched [C14].
Repair-order countermodel	Two accepted repair orders reaching different normal forms under the stated local-diamond and completeness hypotheses kill the consensus mechanism.
Closure uniqueness	A second solution of the P or N closure, or a demonstrated target leak in a declared source map, kills the compression claim.

7. Napkin probability calculation

One source, every major layer

One finite self-reading architecture reaches quantum theory, 3 + 1 spacetime, Einstein gravity, the Standard Model, the weak hierarchy, the charged-lepton spectrum, the strong scale, cosmic closure, and Newton normalization. Sector-specific fitted decimals in this core: 0.

Core event	Generous window	Factor
The same patch rules support Lorentz 3 + 1 kinematics and the Einstein chain	one chance in ten	10^{-1}
The realized compact branch lands on the exact Standard Model quotient, hypercharge/matter lattice, three colors, and three generations	one chance in one thousand	10^{-3}
The electromagnetic source/root value is a branch witness. Its strict dependency certificate is absent, and the empirical external-data closure misses the measured endpoint	excluded from this score	1
The same electroweak branch lands the 10^{-17} hierarchy exponent ($\epsilon_H = 0$) and the strict source-audit W/Z branch to 10^{-3} of measurement, with the criticality top and Higgs at +0.02% and +0.5% and the fit-free relation check counted inside this row	one chance in one thousand	10^{-3}
The same pixel source fixes the strong scale Λ_{QCD} to $\sim 1\%$ of measurement with no hadronic input	one chance in ten	10^{-1}
The gauge integers fix the branch phase and the balanced carrier lands both charged-lepton mass ratios at 9.83 and 30.94 ppm with no lepton-mass input	one chance in one hundred	10^{-2}
The certified lepton mass intervals, the down-type ratio sector, the hadronic companion, the nucleon ratio, and the galaxy scorecard consume declared empirical or external inputs	excluded from this score	1

$$p_{\text{coincidence}} = (10^{-1})(10^{-3})(1)(10^{-3})(10^{-1})(10^{-2}) = 10^{-10}, \quad -\log_2 p_{\text{coincidence}} = 33.2 \text{ bits.}$$

For a napkin calculation, give OPH and a catch-all coincidence model equal prior odds, approximate the five scored clusters as independent, use the product as the likelihood of the evidence under coincidence, and apply a 10^4 search penalty. The resulting score is

$$\Pr(\text{OPH incorrect} \mid E) \approx \frac{10^4 p_{\text{coincidence}}}{1 + 10^4 p_{\text{coincidence}}} = 0.0001\%,$$

about one chance in one million. A calibrated posterior would require measured likelihoods and a defensible prior; this score keeps its assumptions visible. Each factor is deliberately generous by orders of magnitude against the printed offsets: the lepton-ratio cluster alone sits at the ten-parts-per-million scale and is scored at one in a hundred, while rows that consume declared empirical or external inputs contribute nothing to the product. As a severe robustness check, square-rooting every factor leaves 10^{-5} ; after the same search penalty, the coincidence reading still loses at odds of ten to one.

The odds argument in plain language

Picture a key cut from one blueprint. A random key that opens one door proves little; locks give way to luck once in a while. This key opens the door of quantum probability, the door of relativistic spacetime, the exact combination lock of the Standard Model’s discrete architecture, and a row of numerical locks whose combinations run to seven digits, and the blueprint was frozen before the locks were inspected. Under the stated independence approximation, separate strokes of luck multiply, the way two dice showing sixes is rarer than one. The table above scores each lock with deliberately lazy generosity (the two lepton ratios land within ten parts per million and are scored as if the chance were one in a hundred), and the five scored locks multiply out to one in ten billion. Large exponents are no cause for panic: every factor is visible above, and each is lazy in OPH’s disfavor.

Two objections deserve numbers rather than words. First: perhaps thousands of blueprints were tried in private and only the winner is on display. The score pays a ten-thousand-fold search penalty for that possibility. Second: perhaps some locks were opened by copying the combination. Any output whose calculation touched measured data is scored at zero evidence in this budget, however well it lands, the way a weather forecast issued after looking out the window scores nothing. After both deductions, the napkin score is one in a million.

The reading is direct. One finite premise, frozen once, emits the discrete architecture of the Standard Model, the continuous spacetime of general relativity, and a scoreboard of numerical endpoints that measurement keeps confirming, while its recorded rejections show the machinery is willing to lose. Under the assumptions displayed above, coincidence is the losing explanation by six orders of magnitude. The remaining alternatives are the stated open gates, and each carries a registered test. That is the sense in which we most likely inhabit an OPH simulation: the hypothesis is compact, the evidence is overconstrained, and every escape route is a named, checkable computation.

References

Data and standards

- [S1] CODATA Task Group (2024). “2022 CODATA recommended values: Newtonian constant of gravitation.” NIST Reference on Constants.
 - [S2] CODATA Task Group (2024). “2022 CODATA recommended values: inverse fine-structure constant.” NIST Reference on Constants.
 - [S8] Particle Data Group (2026). “Leptons.” Review of Particle Physics summary table.
 - [S9] NuFIT collaboration (2026). “NuFIT 6.1: three-flavour global analysis of neutrino oscillation data.” www.nu-fit.org.
 - [S10] Keshavarzi, A., D. Nomura, and T. Teubner (2020). “ $g - 2$ of charged leptons, $\alpha(M_Z^2)$, and an update of the hadronic contributions.” *Phys. Rev. D* 101, 014029.
 - [S11] Particle Data Group (2026). “Quarks.” Review of Particle Physics summary table.
- OPH papers** All OPH series papers below use release **r1535**.
- [1/O1] Mueller et al. (2026). “Observers Are All You Need.”
 - [2/O2] Müller et al. (2026). “Reality as a Consensus Protocol: The Fixed-Point Computation That Implements Physics.”
 - [3/O3] Müller et al. (2026). “Federated Echoshedral Screen Microphysics: Patch Hardware, Records, and Observer Synchronization in OPH.”
 - [4/O4] Müller et al. (2026). “Recovering Relativity and the Standard Model from Observer Overlap Consistency.”
 - [5/O5] Müller et al. (2026). “Deriving the Particle Zoo from Observer Consistency.”

- [6/O6] Müller, B. (2026). “The Fine-Structure Constant as an OPH Pixel Fixed Point.” OPH technical note.
- [7/O7] Mueller, B., and D. Matscheko (2026). “Observer-Patch Holography and the Dark Matter Phenomenon.”
- [8/O8] Müller, B. (2026). “Explaining the Yang–Mills Mass Gap with Observer-Patch Repair Dynamics.” OPH technical note.
- [9/O9] Müller, B. (2026). “Observer-Patch Holography as a String-Vacuum Selector.” OPH technical note.

Executable artifacts

- [C1] FloatingPragma (2026). Pixel-closure solver and receipts.
- [C2] FloatingPragma (2026). Particle-branch source and audit suite.
- [C3] FloatingPragma (2026). Hierarchy theorem package and receipts.
- [C4] FloatingPragma (2026). Electroweak calibration receipts.
- [C5] FloatingPragma (2026). Consensus schedule benchmarks.
- [C6] FloatingPragma (2026). Lean 4/Mathlib theorem artifact.
- [C7] Mueller, B. (2026). OPH-FPE simulation and receipts, including the quantum-record, BW/KMS, H^3 , and geometry runs.
- [C8] FloatingPragma (2026). Lorentz–Einstein geometry receipts.
- [C9] FloatingPragma (2026). SPARC galaxy scorecard and receipts.
- [C10] FloatingPragma (2026). Charged-lepton interval and orbit-geometry lanes.
- [C11] FloatingPragma (2026). Flavor register lanes and negative scans.
- [C12] FloatingPragma (2026). Declared-empirical hadronic spectral lanes.
- [C13] FloatingPragma (2026). Theorem gap register.
- [C14] FloatingPragma (2026). Falsifiability map and kill conditions.